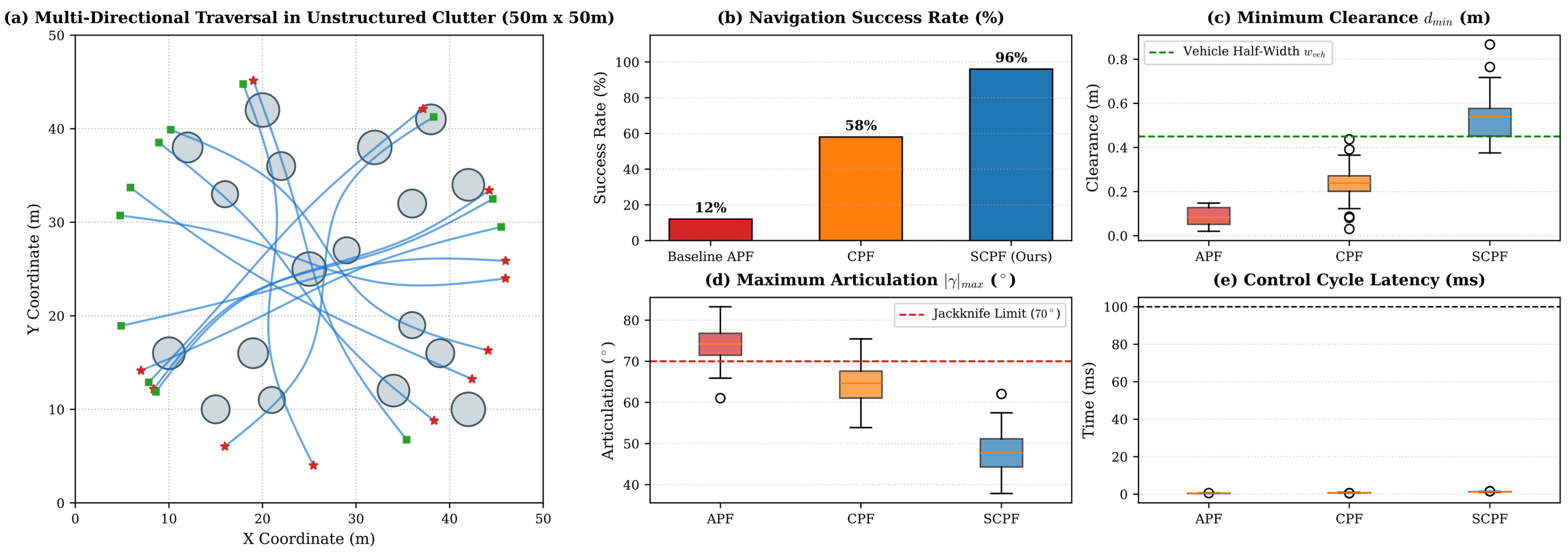

(a) Multi-Directional Traversal in Unstructured Clutter (50m x 50m)
X Coordinate (m)
Y Coordinate (m)
(b) Navigation Success Rate (%)
Success Rate (%)
12%
58%
96%
Baseline APF
CPF
SCPF (Ours)
(c) Minimum Clearance d_{min} (m)
Vehicle Half-Width w_{veh}
Clearance (m)
APF
CPF
SCPF
(d) Maximum Articulation $|\gamma|_{max}$ (°)
Jackknife Limit (70°)
Articulation (°)
APF
CPF
SCPF
(e) Control Cycle Latency (ms)
Time (ms)
APF
CPF
SCPF